

Homework rules

- An office hour for HW0 will be held on 14:00, 3/2, Mon at 資439, where we will teach you how to write the proofs for Prob 4.
- Group discussion is *highly recommended*. But you have to write down your solution *by yourself*. Aim for clarity and conciseness in your solution, emphasizing the main ideas over low-level details.
- You should *cite* all the references that help you to solve the homework, including the people you have discussed with and the online material you looked at.
- You are required to submit a *pdf* by email to csie.tamc@gmail.com. You may type your solution in English (preferable) or Chinese.
- You are not expected to solve *all* problems. Just try your best! If you don't solve a problem, try your best to write down your thinking process.
- You are not allowed to use LLMs in any way to work on your homework.
- 作業總計26+4(bonus)分。

1. **(Working with negligible functions)** A non-negative function $\nu: \mathbb{N} \rightarrow \mathbb{R}$ is negligible if it decreases faster than the inverse of any polynomial; more precisely, for each polynomial P with coefficients in $\mathbb{R}$ (and non-negative leading coefficients), there exists some $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{P(n)}$ whenever $n > N$. Otherwise, we say that ν is non-negligible.

(i) (1 point) Show that ν is negligible if and only if for every fixed sufficiently large integer c we have

$$\lim_{n \rightarrow \infty} \nu(n) \cdot n^c = 0.$$

(The displayed condition can be denoted by $\nu(n) = o(n^{-c})$.)

In the following, $\text{negl}(n)$ denotes some arbitrary negligible function, and $\text{poly}(n)$ denotes some arbitrary polynomial in n with non-negative leading coefficient. State whether each of the following functions is negligible or non-negligible, and give a brief justification.

- (ii) (1 point) $\nu(n) = 1/2^{100 \log n}$.
- (iii) (1 point) $\nu(n) = n^{-\log \log \log n}$.
- (iv) (1 point) $\nu(n) = \text{poly}(n) \cdot \text{negl}(n)$. (State whether ν is always negligible, or not necessarily.)
- (v) (1 points) $\nu(n) = \text{negl}(n)^{\frac{1}{\text{poly}(n)}}$. (Same instructions as previous item.)
- (vi) (2 point) Prove or disprove the following statement: $\nu(n)$ is non-negligible if and only if there exists a polynomial P with coefficients in $\mathbb{R}$ (and non-negative leading coefficients), there exists some $N \in \mathbb{N}$ such that $\nu(n) \geq \frac{1}{P(n)}$ whenever $n > N$.
- (vii) (2 points) Given finitely many tuples $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}$ where n_i 's are all distinct, show that there are always a negligible function ν and a non-negligible function f such that $\nu(n_i) = f(n_i) = y_i$ for each i . Does the same conclusion still hold for infinitely many such tuples?

2. **(Large deviation bounds)** Assume that $X_1, \dots, X_n$ are independent identically distributed (i.i.d.) random variables, each taking 1 with probability p and 0 with probability $1 - p$. The Chernoff's bound says that for all $\epsilon > 0$,

$$\Pr \left[\left| \frac{1}{n} \sum_i X_i - p \right| \geq \epsilon \right] \leq 2e^{-2n\epsilon^2}.$$

If you are rusty on Chernoff's bound, read about it, e.g., [here](#) or search Google; there are lots of forms of the bound, the above being the most convenient for our applications.

- (i) (2 points) How large should n be if we want the average of the X_i to be within $\pm\epsilon$ of p with probability at least $1-\delta$? (asymptotic expression for n is enough)
- (ii) (3 points) Imagine we used Chebyshev's bound instead of Chernoff's, and if you wish, assume for simplicity that $p = 1/2$. What bound on n would you get then? Do you see any advantage of Chebyshev's bound over Chernoff's?

3. **(Union bounds)** Let $A, B_1, \dots, B_k$ be some arbitrary probabilistic events.

- (i) (1 point) Show that

$$\Pr[B_1 \cup B_2 \cup \dots \cup B_k] \leq \sum_{i=1}^k \Pr[B_i].$$

- (ii) (1 point) Show that

$$\Pr[A \cap (B_1 \cup B_2 \cup \dots \cup B_k)] \leq \sum_{i=1}^k \Pr[A \mid B_i].$$

- (iii) (1 point) Let F be the set of functions mapping $\{1, 2, \dots, n\}$ to $\{1, 2, \dots, m\}$, where $m \geq n^2$. Show that

$$\Pr_{f \leftarrow F}[f \text{ is injective}] \geq 1 - \frac{n^2}{2m}.$$

Specifically, we use $f \leftarrow F$ to denote that f is chosen uniformly at random from F and a function is injective if $f(x) \neq f(y)$ for any distinct x, y .

4. **(One-Way Functions)** We first review the concept of one-way functions which we've introduced in the lecture. A function $f: \{0, 1\}^* \rightarrow \{0, 1\}^*$ is one-way if it can be efficiently computed, but no efficient algorithm can invert it. More precisely, for any probabilistic polynomial time (PPT) algorithm Adv ,

$$\Pr_{x \xleftarrow{\$} \{0, 1\}^n} [f(\text{Adv}(1^n, f(x))) = f(x)] = \text{negl}(n)$$

In this problem, we focus on one-way function f that is length-preserving, i.e., for every $x \in \{0, 1\}^*$ it holds that $|f(x)| = |x|$. Note that $|$ denotes concatenation, e.g., $1011|0000 = 10110000$, $\oplus$ denotes bit-by-bit XOR, $\wedge$ denotes bit-by-bit AND, and $x_{[a:b]}$ denotes the substring of x from the a th bit to the b th bits.

For each of the following functions g , assuming f is a one-way function, prove that g is a one-way function, or provide a counter example to demonstrate that it is not always a one-way function. More specifically:

- In the case that g is one-way, you should show a reduction from inverting f to inverting g . In other words, you should show that if g is efficiently invertible, then f is also efficiently invertible.
- In the case that g is not always one-way, you should do the following. Assuming only that length-preserving one-way functions exist, you should construct a length-preserving one-way function f such that when instantiated with f , g is not a one-way function. That is, you should show a polynomial-time inverting algorithm for g .

For example, to prove that the construction of $g(x) = f(x_{[1:n/2]} || 0^{n/2})$ is not always one-way, the argument goes as follows. Assume that h is an arbitrary length-preserving one-way function.

Construct the function

$$f(x) = \begin{cases} 0^n & x_{[n/2+1, n]} = 0^{n/2} \\ h(x) & \text{otherwise.} \end{cases}$$

f is one-way assuming that h is one-way (you have to show this by a reduction). Furthermore, g constructed using this f is not one-way (you have to show this by exhibiting a polynomial-time attack). In general, your solutions should contain complete proofs that the counter examples are valid. In the following, assume $|x| = n$,

- (i) (3 points) $g(x) = f(x_{[1:\frac{n}{2}]})|x_{[\frac{n}{2}+1:n]}$
- (ii) (3 points) $g(x) = f(x)|f(f(x))$
- (iii) (3 points) $g(x) = f(f(x))$ where f is a one-way permutation, i.e. f is one-way *and* bijective.
- (iv) (Bonus 2 points) $g(x) = f(x) \oplus x$
- (v) (Bonus 2 points) $g(x|x') = f(x \wedge x')$ for all $x, x' \in \{0, 1\}^n$. Here $g: \{0, 1\}^{2n} \rightarrow \{0, 1\}^n$